FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Acute Rheumatic Fever & Rheumatic Heart Disease

Tutorial | Week 4 · Small babies, sugar & the heart | Paediatrics | Dr Aliku

Acute rheumatic fever is an autoimmune reaction 2–4 weeks after group A streptococcal sore throat that inflames joints, brain, skin and the heart; Uganda is a high-risk setting, so use the 2015 revised Jones criteria with high-risk thresholds, treat the acute illness and any heart failure, and above all give benzathine penicillin every 4 weeks for years, because only secondary prophylaxis stops rheumatic heart disease progressing.

Must know before the ward round

1. Revised Jones criteria (AHA 2015, still current, endorsed by WHO 2024): initial ARF = evidence of preceding GAS infection plus 2 major, or 1 major + 2 minor. Recurrence = 2 major, 1 major + 2 minor, or 3 minor.
2. Major (J♥NES): Joints (arthritis), Carditis (clinical or subclinical on echo), Nodules (subcutaneous), Erythema marginatum, Sydenham chorea.
3. In high-risk populations like Uganda, MONOarthritis or polyarthralgia can be a major criterion. Minor criteria are monoarthralgia, fever ≥38.0 °C, ESR ≥30 mm/h and/or CRP ≥3.0 mg/dL (30 mg/L), and a prolonged PR interval.
4. Sydenham chorea (and indolent carditis) alone is enough to diagnose ARF, with no other criteria or streptococcal evidence needed.
5. Eradicate GAS at diagnosis: benzathine penicillin G (BPG) IM single dose, 0.6 MU if under 30 kg and 1.2 MU if 30 kg or more (UCG 2023; AHA uses 27 kg), or penicillin V/amoxicillin for 10 days. The same injection starts secondary prophylaxis.
6. Arthritis: aspirin 80–100 mg/kg/day in 3–4 doses (UCG 2023) until symptoms settle, then taper; naproxen 10–20 mg/kg/day in 2 doses (max 1 g/day) is a safer alternative. Arthritis should improve dramatically within 24–72 hours; if not, rethink the diagnosis.
7. Heart failure from carditis: furosemide 1 mg/kg (WHO), spironolactone, an ACE inhibitor (enalapril 0.1 mg/kg/day, titrated) and fluid/salt restriction. Prednisolone 1–2 mg/kg/day is used for severe carditis, though evidence is weak.
8. Secondary prophylaxis: BPG IM every 4 weeks (every 3 weeks if breakthrough or high-risk). Oral penicillin V 250 mg twice daily if injections are impossible; erythromycin if truly allergic.
9. Duration (UCG 2023): no carditis = 5 years or until age 18–21. Carditis without residual disease = 10 years or until age 25. Residual valve disease = until age 40–45 or for life. (AHA 2020: whichever is longer of 5 y/age 21, 10 y/age 21, 10 y/age 40.)
10. GOAL trial (Uganda, NEJM 2022): BPG every 4 weeks in children with latent (echo-detected) RHD cut 2-year progression from 8.2% to 0.8%. WHO 2024 now recommends prophylaxis for screen-detected RHD in children.
11. RHD most often damages the mitral valve (regurgitation in children, then stenosis), then the aortic valve. Apical pansystolic murmur radiating to the axilla = MR; Carey Coombs mid-diastolic murmur = acute mitral valvulitis.
12. WHO 2024: penicillin (IM or oral) is first line for GAS pharyngitis. In high-risk settings without testing, treat clinically suspected GAS sore throat in children with antibiotics.

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1. Definition

Acute rheumatic fever (ARF) is an **autoimmune inflammatory illness** that follows an untreated **group A *Streptococcus* (GAS, *Streptococcus pyogenes*) throat infection** (and possibly skin infection) by about **2–4 weeks**. It mainly affects **children aged 5–15 years**. It causes **migratory arthritis, carditis, chorea, skin rash and nodules**. The joints, brain and skin recover completely, but the **heart valves may scar permanently**. This is **rheumatic heart disease (RHD)**.

Term	Meaning
Acute rheumatic fever	Diagnosed by the 2015 revised Jones criteria (AHA; endorsed by WHO 2024): evidence of preceding GAS infection + 2 major, or 1 major + 2 minor (section 5)
Recurrent ARF	A new episode in a child with previous ARF or RHD; can be diagnosed with 2 major, 1 major + 2 minor, or **3 minor** criteria
Rheumatic heart disease	Permanent valve damage (mainly mitral, then aortic) after one or more episodes of ARF
Subclinical carditis	No murmur, but **pathological mitral or aortic regurgitation on Doppler echocardiography**; counts as carditis (major criterion) since 2015
Latent RHD	Valve changes found by **echo screening** in an asymptomatic child who never had recognised ARF (World Heart Federation criteria)

The disease is **preventable at three levels**:

1. **Primordial**: housing, less crowding, nutrition.
2. **Primary**: treat GAS sore throats with penicillin.
3. **Secondary**: **benzathine penicillin G (BPG) injections every 4 weeks** for years after ARF or RHD diagnosis. This is the single most important intervention you will prescribe.

> **CLINICAL PEARL**
> The one thing never to forget: **ARF "licks the joints but bites the heart."** The arthritis is dramatic and goes away; the carditis is often silent and stays. Every child with ARF or RHD needs an **echocardiogram** and **monthly BPG**, and a missed injection is a chance for a recurrence that damages the valves further.

2. Epidemiology

- **Global**: RHD is the **most common acquired heart disease in people under 25** worldwide (WHO: about **55 million affected, 360,000 deaths a year**), almost all in low- and middle-income countries.
- **Age**: first episode usually at **5–15 years** (UCG: 3–15). Rare under 3. Recurrences can continue into adulthood.

- **Uganda**: a **high-risk setting**. Echocardiographic school screening studies (Kampala, Gulu, Lira and others) found latent RHD in roughly **1.5–3% of school-age children** Verify *district figures*. Most of these children had never been diagnosed with ARF, because ARF often presents as "joint pains" treated at drug shops. Many Ugandan children are first diagnosed when they already have **heart failure from severe mitral or aortic valve disease**.
- **Services**: the **Uganda Heart Institute (UHI) at Mulago** is the national referral centre for echo, catheter interventions (e.g. balloon mitral valvotomy) and open-heart surgery. **RHD clinics** run at several regional referral hospitals (e.g. **Lira, Gulu, Mbarara**) Verify *current list*, linked to a national **RHD registry**. Many children still need surgery abroad, funded by charities.
- **BPG supply**: global stock-outs of benzathine penicillin have affected Uganda. Families travel monthly for injections, so **adherence** is a major problem.
- **Pregnancy**: girls with RHD later face high risks in pregnancy (see the obstetrics topic on cardiac disease in pregnancy). WHO 2024 supports **echo screening in antenatal care** in high-risk settings.

3. Aetiology and Risk Factors

- **Cause**: an autoimmune response to **GAS pharyngitis** (or possibly impetigo or infected scabies) in a genetically susceptible child.
- **Risk factors**: **overcrowding** (school dormitories, shared beds), poverty, poor access to treatment of sore throat, undernutrition, genetic susceptibility (HLA types; familial clustering), female sex for RHD.
- **Risk factors for recurrence and progression**: missed BPG injections (stock-outs, distance, painful injections), repeated GAS exposure in crowded households and boarding schools.
- **Precipitants of heart failure in established RHD**: **ARF recurrence**, infection (pneumonia, **infective endocarditis**), anaemia, atrial fibrillation (AF), pregnancy, stopping medication.

4. Pathophysiology

From sore throat to scarred valve

1. **GAS pharyngitis** (or possibly impetigo) in a genetically susceptible child.
2. **Molecular mimicry**: antibodies and T cells against streptococcal **M protein** and **N-acetylglucosamine** (cell wall carbohydrate) **cross-react** with human proteins: **cardiac myosin**, **laminin and keratin on valve endothelium**, **joint synovium**, and **basal ganglia neurons** (lysoganglioside, tubulin).
3. After a **latent period of 2–4 weeks** (1–5 weeks), inflammation appears in joints, heart, skin and brain. **Chorea** has a longer latency (**1–6 months**), so ASO may have normalised by then.
4. **Carditis is a pancarditis** (endocardium, myocardium, pericardium), but **valvulitis** causes the lasting damage. **Aschoff bodies** (granulomas with Anitschkow "caterpillar" cells) are the histological hallmark.
5. **Repeated attacks** cause more scarring: leaflet thickening, **commissural fusion**, chordal shortening and fusion, leading to **regurgitation** (early, in children) and then **stenosis** (later, but can be severe in African adolescents: "juvenile mitral stenosis").

Valve involvement: **mitral (~65–70%)** > **mitral + aortic (~25%)** > aortic alone > tricuspid (rarely isolated, usually functional from pulmonary hypertension). Pulmonary valve almost never.

Why the heart fails

- **Acute carditis**: severe MR (valvulitis + annular dilatation + chordal elongation) → volume overload → left atrial and pulmonary venous congestion → **pulmonary oedema**. Myocarditis contributes less than once thought.
- **Chronic RHD**: MR → left ventricular dilatation; MS → raised left atrial pressure → **pulmonary hypertension**, **atrial fibrillation**, right heart failure; AR → LV volume overload, wide pulse pressure.

GAS pharyngitis in a susceptible child

↓

Antibodies and T cells to M protein and N-acetylglucosamine

↓

Molecular mimicry: myosin, valve laminin, synovium, basal ganglia

↓

Arthritis, chorea, erythema marginatum, nodules: resolve	Pancarditis, valvulitis
↓	
Acute MR, heart failure	Scarring, commissural fusion
↓	
Recurrent GAS without prophylaxis: more scarring	Chronic RHD: MR, MS, AR

↓

LV dilatation, pulmonary hypertension, AF

↓

Heart failure, stroke, endocarditis, death

↻ From “Recurrent GAS without prophylaxis: more scarring” back to “Scarring, commissural fusion”

Feature links

- Synovial cross-reaction → **migratory, exquisitely painful large-joint arthritis** that heals without damage and responds dramatically to NSAIDs.
- Basal ganglia cross-reaction (long latency) → **Sydenham chorea**, emotional lability, with ASO often already normal.
- Skin inflammation → **erythema marginatum**; connective tissue granulomas → **subcutaneous nodules** (usually with carditis).
- Valvulitis with annular dilatation → **apical pansystolic murmur** (MR); high flow across the inflamed mitral valve → **Carey Coombs mid-diastolic murmur**.
- Pericardial inflammation → **pericardial rub**, effusion, low ECG voltages.
- Myocardial and conduction tissue inflammation → **tachycardia out of proportion to fever** (persisting in sleep), **prolonged PR**, heart block.
- Volume overload from MR or AR → **LV dilatation**, displaced apex, gallop, **pulmonary oedema**.
- Aortic regurgitation → **collapsing pulse**, wide pulse pressure, early diastolic murmur.
- Mitral stenosis → raised LA pressure → **loud S1, opening snap, mid-diastolic rumble**, **haemoptysis**, pulmonary hypertension, **AF** → **stroke and emboli**.
- Damaged valves → bacteraemia seeds them → **infective endocarditis** (fever, changing murmur, splinters, haematuria, splenomegaly).
- Systemic inflammation → **fever, raised ESR and CRP**, anaemia of inflammation, leucocytosis.

5. Classification / Staging

The 2015 revised Jones criteria (AHA; endorsed by WHO 2024)

First decide whether the child comes from a **low-risk** population (ARF incidence ≤2 per 100,000 school-aged children per year, or all-age RHD prevalence ≤1 per 1,000) or a **moderate/high-risk** population (everyone else). **Ugandan children are moderate/high risk.**

	Low-risk populations	Moderate/high-risk (Uganda)
Major	Carditis (clinical and/or subclinical), **polyarthritis only**, chorea, erythema marginatum, subcutaneous nodules	Carditis (clinical and/or subclinical), **monoarthritis or polyarthritis, and/or polyarthralgia** (if other causes excluded), chorea, erythema marginatum, subcutaneous nodules
Minor	Polyarthralgia, fever **≥38.5 °C**, **ESR ≥60 mm/h** and/or CRP ≥3.0 mg/dL, prolonged PR (age-adjusted, unless carditis is a major criterion)	**Monoarthralgia**, fever **≥38.0 °C**, **ESR ≥30 mm/h** and/or **CRP ≥3.0 mg/dL (30 mg/L)**, prolonged PR (age-adjusted, unless carditis is a major criterion)

Diagnosis

Situation	Requirement
Initial ARF	Evidence of preceding GAS infection + **2 major** or **1 major + 2 minor**
Recurrent ARF	Evidence of preceding GAS infection + **2 major**, **1 major + 2 minor**, or **3 minor**
Sydenham chorea or **indolent carditis**	Can be diagnosed **alone**, without other criteria or GAS evidence (they appear months after the infection)

Evidence of preceding GAS infection: raised or rising **antistreptolysin O (ASO)** titre or **anti-DNase B**; positive **throat culture**; positive **rapid GAS antigen test**; recent scarlet fever (WHO Pocket Book).

> UGANDA CONTEXT
> **UCG 2023** prints a simplified Jones table: evidence of strep infection + 2 major or 1 major + 2 minor (polyarthritis listed as major, ESR/CRP and PR prolongation as minor). For exams and the ward, use the **2015 high-risk version**, which the **WHO 2024 guideline** endorses ("the Jones criteria should be used"). It lets you count **monoarthritis and polyarthralgia** as major, which matters because many Ugandan children have already had NSAIDs or herbal remedies before arthritis becomes "poly".

Subclinical carditis and latent RHD (echo criteria)

- **Latent RHD** is defined by the **World Heart Federation (WHF) criteria** (2012, updated 2023), and **WHO 2024** recommends prophylaxis for children meeting minimum criteria.
- **Pathological MR on Doppler (WHF)**: seen in **2 views**, jet length **≥2 cm** in at least one view, peak velocity **≥3 m/s**, and **pansystolic** in at least one envelope.
- **Pathological AR**: 2 views, jet **≥1 cm**, velocity ≥3 m/s, **pandiastolic**.

Heart failure severity

- **Infants and young children**: modified **Ross** classification (feeding, sweating, tachypnoea, growth).
- **Older children**: **NYHA class I–IV** (I: no limitation; II: symptoms on ordinary activity; III: symptoms on less than ordinary activity; IV: symptoms at rest).

Age-adjusted upper limit of the PR interval: 3–12 years **0.16 s**; 12–16 years **0.18 s**; ≥17 years **0.20 s**.

6. Clinical Features

History

Ask about	Why it matters
Sore throat 2–4 weeks ago (or skin sores 2–6 weeks ago), treatment received, scarlet fever, skin sores/scabies	Preceding GAS; many children recall nothing (only ~⅔ do)

Ask about	Why it matters
Joint pain: which joints, **migratory** (one gets better as another flares), swelling, **exquisitely painful**, prior NSAIDs/aspirin, response to painkillers	Classic ARF arthritis is migratory, large joints, very painful, and responds dramatically to NSAIDs (which also mask the migration)
Fever	Minor criterion
Breathlessness, orthopnoea, cough, chest pain, palpitations, swelling of feet/face, reduced exercise tolerance (NYHA)	Carditis, heart failure, pericarditis, AF
Clumsiness, dropping things, handwriting deteriorating, school performance, emotional lability, "restlessness" noticed by teachers	**Sydenham chorea**
Rash, painless lumps	Erythema marginatum, nodules
Previous ARF, known RHD, BPG card (when last, how many missed)	Recurrence; adherence
Penicillin allergy history (what exactly happened)	Most "allergies" are not true allergy; WHO 2024 advice on test doses
Current medicines (aspirin, warfarin)	Toxicity, interactions, INR
Family history of RHD, household crowding, school (boarding?)	Risk and household screening
Feeding, weight loss, growth, immunisation, HIV status	Nutrition and comorbidity; heart failure causes poor growth
Menarche and sexual activity (adolescent girls)	Pregnancy risk with RHD; contraception counselling; warfarin teratogenicity

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children. In heart failure the ABCDE assessment (section 9) comes first. See also the paediatric cardiovascular examination topic.

Introduction, consent and preparation

- Introduce yourself; confirm the child's name, age and the carer's relationship.
- Explain; consent from the parent and assent from the child.
- **Hand hygiene**.
- Position: **sitting at 45°** (upright if orthopnoeic); expose the chest, keeping an adolescent girl's dignity (chaperone).

General inspection (end of the bed)

- Comfortable or distressed; **posture** (sitting up = orthopnoea); breathlessness at rest.
- **Pallor** (anaemia worsens failure); cyanosis; oedema.
- **Involuntary, purposeless, jerky movements** of the face and hands (chorea); emotional lability.
- Nutrition and growth: weight, height, Z-scores.
- **Vital signs**: HR (**tachycardia out of proportion to fever**, persistent in sleep), rhythm, RR, BP (wide pulse pressure in AR), SpO_2, temperature.

Hands, face and neck

- **Hands**: clubbing, **splinter haemorrhages**, **Osler nodes**, **Janeway lesions** (endocarditis); peripheral cyanosis; **chorea signs**: **milkmaid's grip** (squeezes and releases rhythmically), **pronator sign** (arms held up turn palms outward), **spooning** of outstretched hands, hypotonia.
- **Pulse**: rate, rhythm (AF in chronic MS), volume and character (**collapsing** in AR).
- **Face**: pallor, conjunctival petechiae, **darting tongue** ("jack-in-the-box"), dental hygiene (endocarditis risk).

- **Neck**: **JVP**, carotid pulsation (AR).

Joints and skin

- **Joints**: swelling, warmth, redness, **severe tenderness out of proportion**, limited movement; usually knees, ankles, elbows, wrists. Examine each large joint.
- **Skin**: **erythema marginatum** (pink rings with pale centres on the trunk and proximal limbs, non-itchy, evanescent; hard to see on dark skin), **subcutaneous nodules** (painless, pea-sized, over extensor surfaces, occiput and spine; usually with carditis).

Cardiovascular system

- **Inspection**: precordial bulge, visible apex.
- **Palpation**: **apex position and character** (displaced in LV dilatation), heaves (RV heave in pulmonary hypertension), thrills.
- **Auscultation**: each **murmur by timing, site, grade, radiation**; **pericardial rub**; gallop.

Murmur	Lesion	Where and how
Apical pansystolic, radiating to the axilla	**Mitral regurgitation** (commonest in children)	Diaphragm; louder lying on the left side
Apical mid-diastolic (Carey Coombs)	**Acute mitral valvulitis** (high flow across an inflamed valve)	Bell, left lateral position; no opening snap
Apical low-pitched mid-diastolic rumble with presystolic accentuation, loud S1, opening snap	**Mitral stenosis** (chronic)	Bell, left lateral position, after exercise
Early diastolic decrescendo, left sternal edge	**Aortic regurgitation**	Diaphragm, sitting forward in held expiration
Pericardial friction rub	Pericarditis	Scratchy, varies with position

Completing the examination

- **Lungs**: basal crackles, effusion.
- **Abdomen**: liver size (congestion), **splenomegaly** (endocarditis).
- Peripheral oedema; **urine dipstick** (haematuria in endocarditis); **fundi** (Roth spots).
- Neurological: chorea signs; ask the child to write or draw a spiral.
- Throat last.
- Thank, re-dress, **hand hygiene**, summarise with Jones criteria counted and heart failure graded.

Red flags

RED FLAG

- **Heart failure**: tachypnoea, orthopnoea, tachycardia, gallop, hepatomegaly, raised JVP, oedema.
- **New AF or embolic event** (stroke): anticoagulation and urgent cardiology review.
- **Fever + changing murmur + splinter haemorrhages/haematuria/splenomegaly** in a child with RHD: **infective endocarditis**. Take 3 blood cultures before antibiotics.
- **Syncope, chest pain on exertion**: severe aortic stenosis or pulmonary hypertension.
- **Severe MS with haemoptysis or pulmonary oedema**: urgent referral to UHI (balloon valvotomy or surgery).
- **Pregnancy in an adolescent with RHD**: high-risk; refer.

Thinking like a clinician

- **"Joint pains" in a Ugandan school child**: think ARF, but also **sickle cell crisis**, **septic arthritis** (single hot joint, very unwell, needs aspiration), **reactive arthritis**, **juvenile idiopathic arthritis**, **leukaemia** (bone pain, cytopenias), **post-streptococcal reactive arthritis** (non-migratory, poor response to NSAIDs). A dramatic response to aspirin within 1–3 days supports ARF.
- **Arthritis that does not respond** to NSAIDs within 72 hours → rethink.
- **Tachycardia at rest and in sleep, out of proportion to fever** → think carditis even before a murmur appears.
- **Every ARF patient needs an echo**: about a third or more of "no carditis" cases have subclinical carditis, which changes prophylaxis duration.
- **A girl with clumsiness and emotional changes** whose teacher thinks she is "playing the fool" may have **Sydenham chorea**. ASO may be normal by then.
- **Infective endocarditis versus ARF recurrence** in a known RHD child with fever: both give fever, raised CRP and a changing murmur. Always do **blood cultures** before antibiotics.
- **Student mistake**: forgetting to **give BPG at diagnosis** and to **book the next injection**. The prescription is incomplete without a prophylaxis card and a date.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Septic arthritis	Single hot joint, very unwell, cannot bear weight, not migratory	Joint aspiration, blood culture
Sickle cell crisis	Known SCD, bone pain, dactylitis, anaemia, jaundice	Hb electrophoresis, FBC
Post-streptococcal reactive arthritis	Arthritis within 10 days of GAS, non-migratory, poor response to NSAIDs, no carditis	Clinical; echo normal (some still give prophylaxis)
Juvenile idiopathic arthritis	Arthritis >6 weeks, morning stiffness, uveitis, salmon rash (systemic JIA)	Clinical; ANA, RF
Leukaemia	Bone pain, pallor, bruising, organomegaly	FBC and film
Infective endocarditis	Known valve disease, fever, embolic phenomena	Blood cultures, echo
Viral myocarditis / dilated cardiomyopathy	Heart failure without valvulitis, no joint signs	Echo: poor LV function
Congenital heart disease (VSD, congenital MR)	Murmur from infancy, failure to thrive from early life	Echo
Lyme disease, SLE, HSP	Rare here; rash, renal involvement	Specific tests
Other chorea: drugs, SLE, Huntington, Wilson	Family history, drug history	Clinical, specific tests

8. Clinical Workup

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Bedside

Test	Why	Expected finding	Available at
ECG	Minor criterion; rhythm	PR prolongation (above the age limit), heart block, AF, LVH/LA enlargement, low voltages (pericardial effusion)	General Hospital and above
Urine dipstick	Endocarditis; coexisting PSGN	Haematuria (endocarditis); protein (PSGN can coexist)	All levels
HemoCue Hb	Anaemia worsens failure	Low Hb	HC III and above (variable)
Rapid GAS antigen test (if available)	Current GAS pharyngitis	Positive	Rarely available
Handheld or point-of-care echo	WHO 2024 supports **handheld echocardiography** for diagnosing RF carditis and RHD where standard echo is unavailable	Valve regurgitation, thickening	Screening programmes, RHD clinics

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
ESR, CRP	Every suspected ARF; repeat every 1–2 weeks	ESR ≥30 mm/h or CRP ≥30 mg/L (high-risk thresholds); falling with treatment	Minor criterion; guides duration of anti-inflammatory therapy and bed rest	Most hospitals
ASO titre	Every suspected ARF	Raised (above age-specific upper limit; often >200–300 IU/mL in school children) or **rising on repeat after 10–14 days**; peaks 3–6 weeks after infection	Evidence of GAS	Hospitals
FBC	All	Anaemia (of inflammation), leucocytosis	Treat anaemia; exclude leukaemia (cytopenias, blasts)	HC IV and above
U&E, creatinine	Before ACE inhibitors and diuretics, and during	Rising creatinine, low or high K^+	Dose adjustments	Creatinine at hospitals; K^+ at RRH/NRH

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Echocardiography with Doppler	**Every suspected ARF and all known RHD**	Pathological MR/AR (WHF), valve thickening, chordal changes, pericardial effusion, LV dilatation and function, MS gradient, pulmonary pressure	Confirms carditis (including subclinical), stages RHD, decides prophylaxis duration and referral for intervention	UHI, RRHs with cardiac clinics, private centres; handheld echo in programmes
Chest X-ray	Heart failure, chest symptoms	Cardiomegaly, pulmonary venous congestion, LA enlargement (double right heart border, splayed carina), effusion	Severity; alternative diagnoses	Some HC IVs; all hospitals

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Anti-DNase B	Late presentation, after skin infection	Raised (stays raised longer than ASO)	Evidence of GAS	Referral labs
Throat swab culture	Before antibiotics	GAS (positive in only ~10–25% by the time ARF appears)	Evidence of GAS	RRH/NRH
Blood cultures (× 3)	Fever in known RHD, suspected endocarditis	Viridans streptococci, *Staph aureus*, HACEK	Endocarditis: 4–6 weeks IV antibiotics	RRH/NRH
INR	On warfarin	Target 2–3 (AF), 2.5–3.5 (mechanical mitral valve)	Dose adjustment	Limited availability outside referral centres
Salicylate level	High-dose aspirin with tinnitus, vomiting, hyperventilation	Toxic level	Reduce dose or switch to naproxen	Rarely available

Summary: ESR/CRP (inflammation; minor criterion), ASO/anti-DNase B/throat culture (recent GAS; required except for chorea or indolent carditis), ECG (prolonged PR, AF, block), echo (carditis, valve damage; staging and duration of prophylaxis), CXR (heart failure), FBC (anaemia, leukaemia), blood cultures (endocarditis in known RHD with fever).

Putting it together

Model one-liner: "A 12-year-old girl from a boarding school in Lira with **migratory polyarthritis** of the knees and ankles for 10 days, fever 38.4 °C, a **new apical pansystolic murmur** radiating to the axilla, **ESR 78**, **ASO 640 IU/mL** and echo showing **moderate pathological MR**: **initial acute rheumatic fever with carditis** (2 major + 2 minor criteria with GAS evidence)."

Problem list

1. Initial acute rheumatic fever (migratory polyarthritis, carditis).
2. Moderate mitral regurgitation; heart failure status (NYHA class).
3. Need for GAS eradication and secondary prophylaxis.
4. Household and school crowding; adherence planning.

Model case presentation (ward round)

> "**Akello is an 11-year-old girl weighing 27 kg** (weight-for-age Z −1.2, height-for-age Z −0.6, BMI-for-age Z −1.3) from a boarding primary school in Lira. She is **immunised per UNEPI**, eats 3 meals a day at school, and is **HIV-negative**. She presents with **2 weeks of joint pains that moved from the right knee to the left ankle and then the right elbow**, with fever, after a sore throat about 3 weeks ago for which she got no treatment. She has had **exertional breathlessness for 4 days**. On examination: temperature 38.3 °C, HR 128 (also 120 while asleep), RR 30, BP 98/56, SpO_2 96%. Her left ankle is swollen, warm and very tender. There is an **apical grade 3/6 pansystolic murmur radiating to the axilla** and a soft apical mid-diastolic murmur, no gallop, and the liver is 2 cm. **ESR 85 mm/h, CRP 96 mg/L, ASO 800 IU/mL, PR interval 0.18 s.** Echo shows **moderate MR with thickened anterior mitral leaflet tips**. Using the 2015 high-risk Jones criteria she has **2 major criteria** (polyarthritis, carditis) and **2 minor criteria** (fever, raised ESR/CRP; the long PR does not count because carditis is already a major criterion), with GAS evidence (ASO): **initial ARF**

> **with moderate carditis**. Plan: **BPG 0.6 MU IM today** (she is under 30 kg), **aspirin 80–100 mg/kg/day in 3–4 doses** (or naproxen), monitor for heart failure, register for **4-weekly BPG for at least 10 years or until age 25** (UCG; lifelong if residual disease), and counsel the family."

Worked example: applying Jones (high-risk)

Child	Findings	Major	Minor	GAS evidence	Verdict
A, 9 y	Swollen painful right knee (monoarthritis), fever 38.2, CRP 60 mg/L, ASO high	Monoarthritis (1)	Fever, CRP (2)	Yes	**Initial ARF** (1 major + 2 minor)
B, 11 y	Polyarthralgia, ESR 45, PR 0.20 s; echo normal; ASO high	Polyarthralgia (1)	ESR, PR (2)	Yes	**ARF** (high-risk rules), if other causes excluded
C, 10 y	Jerky movements, emotional lability, ASO normal, echo shows mild MR	Chorea (alone is enough) + subclinical carditis	None needed	Not required	**ARF (chorea) with carditis**
D, 14 y, known RHD	Fever 38.3, monoarthralgia, ESR 50; ASO high	0	3	Yes	**Recurrent ARF** (3 minor allowed for recurrence)

9. Management

Immediate / emergency

First 5 minutes (ABCDE)

1. **A and B**: respiratory distress, SpO_2; sit the child up if orthopnoeic; **oxygen** if SpO_2 <90% (or <94% with emergency signs) (WHO: aim >90%).
2. **C**: HR (age-adjusted), rhythm, BP, CRT, liver size, JVP. **Signs of heart failure** → start **furosemide 1 mg/kg IV** (WHO); restrict IV fluids.
3. **D**: conscious level; glucose if unwell.
4. **E**: temperature, **Hb (HemoCue)** (anaemia worsens failure).
5. **ECG** if the pulse is irregular or very fast (AF, heart block).
6. If fever + known RHD + new embolic signs or splinters: **blood cultures × 3 before antibiotics** (endocarditis).

Heart failure: general measures: sit up (semi-recumbent), oxygen if SpO_2 <90% or distressed, restrict IV fluids, salt restriction, treat fever and anaemia (transfuse packed cells slowly with furosemide if Hb <7–8 g/dL; WHO Pocket Book says transfuse if Hb <8 g/dL in rheumatic heart failure), treat infection, weigh daily.

Definitive

Acute rheumatic fever: step by step

1. **Admit** (UCG: HC IV and above; WHO: admit) for diagnosis, echo and education. Bed rest while inflammation and failure are active; then **gradual mobilisation** as CRP falls and symptoms settle (strict prolonged bed rest is no longer advised).
2. **Eradicate GAS now**, even if the throat swab is negative, with **one dose of BPG**, which also counts as the **first prophylactic dose**.
3. **Treat arthritis and fever** with aspirin or naproxen (do not start before the diagnosis is clear, if possible, because they hide the migratory pattern; paracetamol is fine while you observe).
4. **Echo** for every child.

5. **Treat heart failure** if present (drug table).
6. **Treat chorea** if disabling (calm environment; valproate or carbamazepine).
7. **Start secondary prophylaxis**, give the family a **BPG card**, book the next injection, and register the child in the **RHD register/clinic**.
8. **Educate**: what ARF is, why injections every 4 weeks matter, signs of heart failure, dental care, contraception and pregnancy planning for girls.

DRUG DOSING

BPG technique: use the **correct dose for weight**, warm the vial to room temperature, inject **slowly (over 2–3 minutes)** deep IM, and consider **lidocaine as diluent** (WHO 2024 conditional recommendation) to reduce pain. Applying pressure or ice beforehand and distraction help. Observe the child for **15–30 minutes** afterwards. Children with **severe valve disease** (severe MS/AR, pulmonary hypertension, poor LV function) can collapse after a **vasovagal** reaction to the painful injection, which is often mistaken for anaphylaxis: give it **lying down**, with an experienced clinician nearby, or discuss oral prophylaxis with cardiology.

Drug table

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Benzathine penicillin G (eradication and prophylaxis)	**0.6 MU (600,000 IU) if <30 kg; 1.2 MU if ≥30 kg** (UCG 2023; AHA cut-off 27 kg); adult 1.2 MU	**Deep IM** (gluteal or lateral thigh)	Single dose; then **every 4 weeks** (3 weeks if breakthrough)	Prophylaxis duration table below	Preferred; counts as first prophylactic dose; give lying down in severe valve disease; lidocaine diluent
or **Penicillin V** (eradication)	**Child 250 mg** (UCG lists 125 mg/dose for young children; 500 mg if ≥30 kg)	Oral	6–12-hourly (UCG 6-hourly)	**10 days**	Adherence for 10 days is poor
or **Amoxicillin** (eradication, sore throat)	**50 mg/kg once daily (max 1 g)**	Oral	Once daily	**10 days**	Good alternative (AHA)
Penicillin allergy: **erythromycin**	10–12.5 mg/kg (max 500 mg)	Oral	6-hourly	10 days	Or **azithromycin** 12 mg/kg once daily (max 500 mg) for 5 days
Aspirin (arthritis, fever)	**UCG 2023: 80–100 mg/kg/day in 3 doses** (WHO Pocket Book: 20 mg/kg 6-hourly for 1–2 weeks, then 15 mg/kg 6-hourly for 3–6 more weeks); **max about 4 g/day**	Oral, with food	6–8-hourly	Until joints settle and CRP falls (usually **2–6 weeks**; UCG 4–8 weeks), then taper	Toxicity: tinnitus, vomiting, hyperventilation; check salicylate level if available. Reye syndrome risk with influenza/varicella. UCG adds an antacid
or **Naproxen**	**10–20 mg/kg/day in 2 doses (max 1 g/day)**	Oral	12-hourly	As above	Preferred in some guidelines (Australia 2020): fewer side effects, twice daily
or **Ibuprofen**	10 mg/kg (max 400–600 mg)	Oral	8-hourly	As above	Alternative

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Prednisolone (severe carditis / heart failure; optional, specialist)	**1–2 mg/kg/day (max 60–80 mg)** (WHO Pocket Book: 1 mg/kg/day for 1 week in severe heart failure)	Oral	Once daily	1–3 weeks, then **taper by ~25% per week**	Weak evidence; WHO 2024 makes **no recommendation** for or against anti-inflammatories to prevent RHD. Overlap with aspirin during taper to avoid rebound
Sodium valproate (disabling chorea)	**10–20 mg/kg/day** (UCG), up to 30 mg/kg/day in 2–3 doses	Oral	8–12-hourly	Until controlled for several weeks, then taper	Or **carbamazepine** 7–20 mg/kg/day in 2–3 doses. Haloperidol is an alternative (extrapyramidal effects). Calm environment
Furosemide (acute failure)	**1 mg/kg**; if no response in 2 h, **2 mg/kg** (WHO Pocket Book). UCG acute: 0.5–1.5 mg/kg/dose; max **6 mg/kg/day** (UCG)	IV (faster) or oral	6–12-hourly as needed, then 1–2 mg/kg once daily maintenance	While congested	Monitor K^+, urine output, weight
Spironolactone	**1–3 mg/kg/day** (UCG 1.5–3 mg/kg/day); max 100 mg/day	Oral	Once or twice daily	Long term	K^+-sparing; check K^+ with ACE inhibitor
Potassium chloride (furosemide alone >5 days, or with digoxin)	**3–5 mmol/kg/day** (WHO Pocket Book)	Oral	Divided	While on furosemide alone	Not needed if on spironolactone or ACE inhibitor
Enalapril	Start **0.05–0.1 mg/kg/day**, titrate over 2 weeks to **0.5 mg/kg/day** (UCG range 0.1–1 mg/kg/day); max **20 mg/day** in children (40 mg adults)	Oral	Once or twice daily	Long term	For MR/AR with LV dilatation. **Avoid in severe AS/MS-dominant disease with low BP.** Check creatinine and K^+. Teratogenic
or **Captopril**	**0.1–0.3 mg/kg/dose** (start low), titrate; max 6 mg/kg/day (max 150 mg/day)	Oral	8-hourly	Long term	Test dose with BP monitoring
Digoxin (AF with fast rate, or poor LV function)	Loading **15 µg/kg** oral, then **5 µg/kg** starting 6 h later (WHO Pocket Book); maintenance **5 µg/kg**; max **250 µg per dose**	Oral	Maintenance **12-hourly**	Long term	Toxic with hypokalaemia; check pulse before each dose (hold if bradycardic); no loading if already on digoxin
Carvedilol (stable chronic failure, specialist)	0.05 mg/kg titrated to 0.35 mg/kg (UCG); max 25 mg/dose	Oral	12-hourly	Long term	Only when euvolaemic and stable
Warfarin (AF, LA thrombus, mechanical valve, prior embolism)	Specialist dosing	Oral	Daily	INR 2–3 (AF); 2.5–3.5 (mechanical mitral)	INR monitoring is difficult outside referral centres; drug interactions; teratogenic in the first trimester

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Amoxicillin (endocarditis prophylaxis, highest-risk only)	**50 mg/kg (max 2 g)**	Oral	Once, 30–60 min before the procedure	Single dose	Clindamycin 20 mg/kg (max 600 mg) or azithromycin 15 mg/kg (max 500 mg) if allergic or already on long-term penicillin

UGANDA CONTEXT

At HC IV/general hospital level you can start **furosemide, spironolactone, an ACE inhibitor (enalapril or captopril), BPG and digoxin** (UCG levels of care HC4/H). **Echo, warfarin monitoring and referral for intervention** happen at RRH cardiac clinics and the **Uganda Heart Institute**. Refer any child with severe valve disease, heart failure despite treatment, AF, or suspected endocarditis.

Referral criteria

Level	What to do	Refer up when
HC III	Treat GAS sore throat (BPG or amoxicillin); give monthly BPG to registered children; recognise heart failure	Suspected ARF, heart failure, new murmur → HC IV or hospital
HC IV / General Hospital	Admit ARF; BPG; aspirin or naproxen; furosemide, spironolactone, ACE inhibitor, digoxin; ECG, ESR, ASO	Echo needed, heart failure not controlled, AF, suspected endocarditis → RRH
RRH (cardiac / RHD clinic)	Echo, warfarin and INR, blood cultures, register	Moderate–severe valve disease, need for valvotomy or surgery, uncontrolled failure, pregnancy with RHD → UHI
NRH: Uganda Heart Institute	Balloon mitral valvotomy, valve repair or replacement, specialist follow-up	Surgery abroad where needed

Level split adapted from UCG 2023 levels of care Verify *UCG level codes*.

Refer to UHI: moderate–severe valve disease, heart failure not controlled, AF or embolism, suspected endocarditis, need for valvotomy or surgery, pregnancy with RHD.

Monitoring and follow-up

- **In hospital**: HR (including asleep), RR, temperature, weight daily; joints; signs of failure; **ESR/CRP every 1–2 weeks**; K^+ and creatinine on diuretics and ACE inhibitors.
- **Aspirin**: watch for tinnitus, vomiting, hyperventilation; taper when joints settle and CRP falls.
- **After discharge**: **BPG every 4 weeks** with the card checked at every visit; trace defaulters; periodic echo; INR on warfarin; growth.
- **Latent RHD on prophylaxis**: re-echo at 2 years and decide (GOAL).

Secondary prophylaxis (the most important step)

Regimen	Dose	Frequency	Notes
Benzathine penicillin G (first line; WHO 2024 strong recommendation)	**0.6 MU if <30 kg; 1.2 MU if ≥30 kg** (UCG 2023)	**Every 4 weeks** (WHO, UCG, AHA); **every 3 weeks** if breakthrough ARF on 4-weekly dosing or very high-risk (AHA)	About **10 times more effective** than oral penicillin for preventing recurrence (Cochrane). Keep a card; trace defaulters

Regimen	Dose	Frequency	Notes
Penicillin V (if injections not possible)	**250 mg** (UCG child 125–250 mg; adolescents 500 mg)	**12-hourly**, daily	Less effective; adherence is hard. WHO 2024: acceptable alternative
Erythromycin (true penicillin allergy)	**250 mg** (UCG child: 10 mg/kg)	**12-hourly**	For anaphylaxis or severe allergy history. WHO 2024: for mild allergy history, consider an **oral penicillin test dose** before BPG; **routine skin testing is not recommended** in those without allergy history

Duration of prophylaxis

Category	UCG 2023	AHA/ACC 2020
ARF **without carditis**	5 years or until age **18–21**	5 years or until age 21, **whichever is longer**
ARF **with carditis but no residual valve disease**	10 years or until age **25**	10 years or until age 21, whichever is longer
ARF **with carditis and residual valve disease** (RHD)	Until age **40–45 or for life**	10 years or until age 40, whichever is longer (sometimes lifelong)
Latent / screen-detected RHD (children)	Not covered	WHO 2024: give prophylaxis (GOAL: 4-weekly BPG; re-echo at 2 years and decide)
After valve surgery	For life (UCG: RHD prophylaxis for life)	Continue per category above

EXAM FAVOURITE

GOAL trial (Beaton et al., *NEJM* 2022; Ugandan children aged 5–17 with **latent RHD**): BPG every 4 weeks for 2 years reduced echo progression from **8.2% to 0.8%**. It is the evidence behind WHO 2024's recommendation to give prophylaxis to children with screen-detected RHD.

Improving adherence: pain reduction (lidocaine, slow injection), reminder systems, cards, community health worker tracing, giving injections close to home, and education of the child and school.

Infective endocarditis prevention and dental care

- **Good dental hygiene** and regular dental review matter more than antibiotic prophylaxis.
- **AHA**: antibiotic prophylaxis before dental procedures that manipulate the gums is now advised only for the **highest-risk** patients: **prosthetic valves or prosthetic material for valve repair, previous endocarditis**, certain congenital heart disease and transplant valvulopathy. Uncomplicated RHD alone is no longer an indication (AHA), though some programmes in endemic countries still give it.
- Regimen: **amoxicillin 50 mg/kg (max 2 g) orally 30–60 minutes before** the procedure (clindamycin 20 mg/kg, max 600 mg, or azithromycin 15 mg/kg, max 500 mg, if allergic). **Note**: a child on long-term penicillin prophylaxis may carry penicillin-resistant mouth streptococci, so the AHA suggests a non-penicillin agent (clindamycin or azithromycin) for these children (specialist advice).

Interventions for chronic RHD (UHI)

Lesion	Intervention
Severe MS with pliable valve	**Percutaneous balloon mitral valvotomy**

Lesion	Intervention
Severe MR/AR with symptoms or LV dilatation/dysfunction	**Valve repair** (preferred in children) or **replacement** (mechanical needs lifelong warfarin; bioprosthetic valves degenerate fast in the young)
Any surgery	Continue BPG prophylaxis; anticoagulation clinic; family planning (warfarin is teratogenic)

10. Complications

Disease-related

Acute	Chronic
Heart failure, pulmonary oedema	**Chronic RHD**: MR, MS, AR, AS, mixed lesions
Pericardial effusion (tamponade rare)	Atrial fibrillation, stroke and systemic emboli
Heart block	Pulmonary hypertension, right heart failure
Death from fulminant carditis (rare)	**Infective endocarditis**
Disabling chorea, emotional disturbance (usually resolves in weeks to months)	Recurrence of ARF with further valve damage
	High maternal mortality in pregnancy; growth failure; school absence; premature death

Treatment-related

Complication	Cause	Recognition and action
Vasovagal collapse (mistaken for anaphylaxis)	Painful BPG injection in severe valve disease	Give lying down, observe 15–30 min, experienced clinician nearby
Salicylate toxicity; Reye syndrome	High-dose aspirin	Tinnitus, vomiting, hyperventilation: reduce dose or switch to naproxen; avoid in influenza/varicella
Rebound inflammation	Stopping prednisolone abruptly	Taper by ~25% per week with aspirin overlap
Hypokalaemia, digoxin toxicity	Furosemide without K^+ sparing; digoxin	Check K^+; add KCl or spironolactone; check pulse before digoxin
Hypotension, rising creatinine, hyperkalaemia	ACE inhibitor (especially in MS-dominant disease)	Start low, check creatinine and K^+; avoid in severe AS/MS with low BP
Bleeding; fetal harm	Warfarin	INR monitoring; contraception and pregnancy planning
Extrapyramidal effects	Haloperidol for chorea	Prefer valproate or carbamazepine

11. Prognosis

- The **joints, brain and skin recover completely**; **chorea** usually resolves in weeks to months.
- **Carditis** determines the outcome: mild regurgitation may regress with strict secondary prophylaxis; **each recurrence adds scarring**.
- **Latent RHD** progresses in about 8% over 2 years without prophylaxis and under 1% with 4-weekly BPG (GOAL).
- Chronic RHD leads to heart failure, AF, stroke, endocarditis, high maternal mortality in pregnancy and premature death, especially when diagnosed late with severe valve disease.

12. Prevention

- **Primordial**: better housing, less crowding (school dormitories, shared beds), nutrition, health education in schools.
- **Primary (the sore throat)**: **WHO 2024**: treat **confirmed GAS pharyngitis** with antibiotics (strong); in **moderate/high-risk settings without testing**, treat **clinically suspected GAS pharyngitis in children and adolescents** (strong); **penicillin (IM or oral)** is first line. In Uganda: **BPG single dose** (0.6 MU <30 kg, 1.2 MU ≥30 kg) or **amoxicillin 50 mg/kg once daily (max 1 g) for 10 days** or penicillin V for 10 days. Treat household contacts with sore throats.
- **Secondary**: **BPG every 4 weeks** for the full duration (section 9); RHD register and defaulter tracing.
- **Screening**: WHO 2024 supports **echo screening of children 5–19** in high-prevalence settings, prophylaxis for screen-detected RHD, and **echo screening in antenatal care**.
- **Tertiary**: heart failure treatment, anticoagulation, endocarditis prevention and dental care, timely valvotomy or surgery, family planning.

13. Special Populations

- **Adolescent girls with RHD**: pregnancy is high-risk (high maternal mortality); contraception counselling; **warfarin is teratogenic** in the first trimester; ACE inhibitors are teratogenic; refer any pregnancy to a high-risk clinic.
- **Severe valve disease (severe MS/AR, pulmonary hypertension, poor LV function)**: give BPG **lying down** with an experienced clinician nearby; avoid ACE inhibitors in MS-dominant disease with low BP.
- **Penicillin allergy**: most "allergies" are not true allergy; WHO 2024: oral penicillin test dose for mild histories; erythromycin for true anaphylaxis.
- **Children on long-term penicillin needing dental procedures**: use clindamycin or azithromycin for endocarditis prophylaxis (resistant mouth streptococci).
- **Under 3 years**: ARF is rare; think congenital heart disease or viral myocarditis first.

14. Key Points

High-yield points

- ARF "**licks the joints but bites the heart**".
- Ugandan children are **high-risk**: monoarthritis or polyarthralgia can be major; fever ≥38.0 °C and ESR ≥30 are minor.
- **Chorea alone** (or indolent carditis) is enough to diagnose ARF.
- **Echo every child with ARF**: subclinical carditis counts and lengthens prophylaxis.
- **BPG at diagnosis** (0.6 MU <30 kg, 1.2 MU ≥30 kg) is also the first prophylactic dose.
- **BPG every 4 weeks** is the most important intervention; oral penicillin is about 10 times less effective.
- Prophylaxis **"5–10–40"** (UCG: 18–21, 25, 40–45 or life).
- **GOAL trial**: progression 8.2% → 0.8% with BPG in latent RHD.
- Mitral regurgitation is the commonest lesion in children; Carey Coombs = acute valvulitis.
- Fever in a child with RHD: **3 blood cultures before antibiotics**.

Numbers & doses

Item	Value
Latent period after GAS pharyngitis	2–4 weeks (chorea 1–6 months)
Peak age	5–15 years
High-risk minor thresholds	Fever ≥38.0 °C; ESR ≥30 mm/h; CRP ≥3.0 mg/dL (30 mg/L)
Low-risk minor thresholds	Fever ≥38.5 °C; ESR ≥60 mm/h
PR upper limit (3–12 y / 12–16 y / ≥17 y)	0.16 / 0.18 / 0.20 s
WHF pathological MR	2 views, jet ≥2 cm, ≥3 m/s, pansystolic
BPG dose (UCG)	0.6 MU <30 kg; 1.2 MU ≥30 kg, deep IM
BPG frequency	Every 4 weeks (3 weeks if breakthrough)
Amoxicillin for GAS sore throat	50 mg/kg once daily (max 1 g) × 10 days
Aspirin (UCG)	80–100 mg/kg/day in 3 doses
Naproxen	10–20 mg/kg/day in 2 doses (max 1 g/day)
Prednisolone (severe carditis)	1–2 mg/kg/day (max 60–80 mg), taper
Furosemide (WHO)	1 mg/kg, then 2 mg/kg if no response
Enalapril	Start 0.05–0.1 mg/kg/day; up to 0.5 mg/kg/day (max 20 mg)
Digoxin (WHO)	Load 15 µg/kg; maintain 5 µg/kg 12-hourly (max 250 µg/dose)
INR targets	2–3 (AF); 2.5–3.5 (mechanical mitral)
GOAL trial progression	0.8% (BPG) vs 8.2% (none) at 2 years
WHO global burden	~55 million with RHD; ~360,000 deaths/year

Exam traps

- **Trap**: Requiring polyarthritis for a Ugandan child. **Right**: in high-risk populations monoarthritis and polyarthralgia can count as major.
- **Trap**: "ASO is normal, so it can't be ARF" in a child with chorea. **Right**: chorea alone is diagnostic.
- **Trap**: Starting aspirin before the diagnosis is clear. **Right**: NSAIDs mask migratory arthritis; use paracetamol while observing (if the child is stable).
- **Trap**: Treating ARF and forgetting secondary prophylaxis. **Right**: give BPG at diagnosis and book the next one in 4 weeks.
- **Trap**: "Oral penicillin is as good as BPG." **Right**: BPG is about 10 times more effective; oral is a fallback.
- **Trap**: Labelling a fainting child after BPG as anaphylaxis. **Right**: in severe valve disease it is often vasovagal; give BPG lying down and observe.
- **Trap**: Giving ACE inhibitors to a hypotensive child with severe mitral stenosis. **Right**: MS needs rate control and diuretics, then mechanical relief.
- **Trap**: Counting a long PR as a minor criterion when carditis is already a major criterion. **Right**: it does not count twice.

Questions seniors ask

Viva questions

1. **What changed in the 2015 Jones criteria?** Stratification into low-risk and moderate/high-risk populations; **subclinical carditis on echo** counts as major; in high-risk populations **monoarthritis and polyarthralgia**

count as major, with lower fever (≥38.0) and ESR (≥30) thresholds for minor criteria; **3 minor** can diagnose recurrence.

2. **Why is ASO often normal in chorea?** Chorea appears 1–6 months after GAS infection, when ASO has fallen; chorea alone is diagnostic.
3. **Which valve is most often affected and how does it present in children?** The mitral valve, usually as **regurgitation** (apical pansystolic murmur to the axilla).
4. **What is the most effective intervention to prevent RHD progression?** Regular **BPG every 4 weeks** (secondary prophylaxis).
5. **How long do you give prophylaxis to a child with carditis that resolved?** UCG: 10 years or until age 25; AHA: 10 years or until 21, whichever is longer.
6. **What is a Carey Coombs murmur?** A soft apical mid-diastolic murmur from high flow across an inflamed mitral valve in acute carditis (not true stenosis).

On the ward round

1. **What is the evidence of streptococcal infection you accept?** Raised/rising ASO or anti-DNase B, positive throat culture or rapid antigen test, recent scarlet fever.
2. **Why do we echo every child with ARF?** To detect **subclinical carditis**, which is a major criterion and lengthens prophylaxis.
3. **How does molecular mimicry explain ARF?** Antibodies to streptococcal M protein and N-acetylglucosamine cross-react with cardiac myosin, valve laminin, synovium and basal ganglia.
4. **What did WHO's 2024 guideline add?** Endorses Jones criteria; handheld echo for diagnosis; echo screening of children 5–19 in high-prevalence settings; prophylaxis for screen-detected RHD; BPG first line; oral penicillin acceptable alternative; lidocaine diluent; adherence support; no recommendation on anti-inflammatories.
5. **When do you refer to the Uganda Heart Institute?** Moderate–severe valve disease, heart failure not controlled, AF or embolism, suspected endocarditis, need for valvotomy or surgery, pregnancy with RHD.
6. **How would you improve adherence to BPG?** Pain reduction (lidocaine, slow injection), reminder systems, cards, community health worker tracing, giving injections close to home, and education of the child and school.

Mnemonics

- **Major criteria "J♥NES"**: **J**oints, ♥ (carditis), **N**odules, **E**rythema marginatum, **S**ydenham chorea.
- **Minor criteria "CAFE PAL"** (classic): **C**RP, **A**rthralgia, **F**ever, **E**SR, **P**rolonged PR, **A**SO (supporting evidence, not a minor criterion), **L**eucocytosis (WHO Pocket Book lists it). Know that the 2015 version uses fever, arthralgia, ESR/CRP and PR.
- **"Licks the joints, bites the heart"** (Lasègue).
- **Prophylaxis duration "5–10–40"**: no carditis 5 years; carditis without residual disease 10 years; residual valve disease until about 40 (or for life).
- **BPG "30 kg divides 0.6 and 1.2"** (UCG).

OSCE tips

- **Chorea station**: demonstrate **milkmaid's grip, pronator sign, darting tongue** and ask the child to write or draw a spiral.
- **Counselling station on BPG**: explain why, how often, how long, what to do if a dose is missed, pain relief, and that **the injection protects the heart**.
- **Cardiovascular station**: present each murmur by timing, site, grade and radiation, name the lesion, and state the heart failure class and the prophylaxis plan.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What is acute rheumatic fever?	An **autoimmune** inflammatory illness about **2–4 weeks after group A streptococcal pharyngitis** (± skin infection), affecting joints, heart, brain, skin and subcutaneous tissue; mainly ages **5–15**.
2	Jones major criteria (2015)?	**J♥NES**: Joints (arthritis), **Carditis** (clinical or subclinical), **Nodules** (subcutaneous), **Erythema marginatum**, **Sydenham chorea**.
3	Minor criteria in moderate/high-risk populations (e.g. Uganda)?	**Monoarthralgia**, **fever ≥38.0 °C**, **ESR ≥30 mm/h and/or CRP ≥3.0 mg/dL (30 mg/L)**, **prolonged PR** (age-adjusted; not if carditis is a major criterion).
4	What joint findings count as MAJOR in high-risk populations?	**Monoarthritis or polyarthritis**, and/or **polyarthralgia** (if other causes excluded). In low-risk populations only polyarthritis counts.
5	Diagnostic rule for initial vs recurrent ARF?	GAS evidence + **initial**: 2 major, or 1 major + 2 minor. **Recurrent**: 2 major, 1 major + 2 minor, or **3 minor**.
6	Which manifestations diagnose ARF on their own?	**Sydenham chorea** and **indolent (insidious) carditis**: no other criteria or streptococcal evidence needed.
7	Acceptable evidence of preceding GAS infection?	Raised or rising **ASO** or **anti-DNase B**, positive **throat culture**, positive **rapid GAS antigen test**, recent **scarlet fever**.
8	Age-adjusted upper limit of the PR interval?	**3–12 y: 0.16 s; 12–16 y: 0.18 s; ≥17 y: 0.20 s.**
9	WHF Doppler criteria for pathological mitral regurgitation?	Seen in **2 views**, jet **≥2 cm** in at least one view, peak velocity **≥3 m/s**, **pansystolic** in at least one envelope.
10	Mechanism of ARF?	**Molecular mimicry**: antibodies/T cells against streptococcal **M protein** and **N-acetylglucosamine** cross-react with **cardiac myosin, valve laminin, synovium and basal ganglia**.
11	Which valves does RHD affect, in order?	**Mitral** (most; regurgitation in children, stenosis later) > **mitral + aortic** > aortic alone > tricuspid (rarely isolated) > pulmonary (almost never).
12	What is a Carey Coombs murmur?	A soft **apical mid-diastolic** murmur in **acute mitral valvulitis** (high flow over an inflamed valve), without an opening snap.
13	Sydenham chorea: signs?	Involuntary jerky movements, **milkmaid's grip**, **darting tongue**, **pronator sign**, spooning of hands, hypotonia, **emotional lability**; absent in sleep. Appears 1–6 months after GAS infection.
14	BPG dose for a child (UCG 2023)?	**0.6 MU (600,000 IU) if <30 kg; 1.2 MU if ≥30 kg**, deep IM. (AHA uses a 27 kg cut-off.)
15	How often is BPG given for secondary prophylaxis?	**Every 4 weeks** (UCG, WHO 2024, AHA); **every 3 weeks** if breakthrough ARF or very high risk (AHA).
16	Duration of prophylaxis (UCG 2023)?	No carditis: **5 years or until age 18–21**. Carditis, no residual disease: **10 years or until 25**. Residual valve disease: **until age 40–45 or for life**.
17	Duration of prophylaxis (AHA/ACC 2020)?	No carditis: 5 y or age 21; carditis without residual disease: 10 y or age 21; carditis with residual disease: 10 y or age 40 — **whichever is longer**.
18	What did the GOAL trial show?	In Ugandan children 5–17 years with **latent RHD**, BPG every 4 weeks for 2 years reduced echo progression from **8.2% to 0.8%** (NEJM 2022).

#	Question	Answer
19	Aspirin dose in ARF arthritis (UCG 2023 and WHO Pocket Book)?	UCG: **80–100 mg/kg/day in 3 doses**. WHO: **20 mg/kg 6-hourly** for 1–2 weeks, then 15 mg/kg 6-hourly for 3–6 weeks. Max about 4 g/day; watch for tinnitus and hyperventilation.
20	Alternative NSAID to aspirin in ARF and its dose?	**Naproxen 10–20 mg/kg/day in 2 divided doses (max 1 g/day)**; ibuprofen also used.
21	Arthritis not improving within 72 hours of NSAIDs: what does it suggest?	Reconsider the diagnosis: septic arthritis, post-streptococcal reactive arthritis, JIA, SCD, leukaemia.
22	Heart failure drugs in rheumatic carditis (with doses)?	**Furosemide 1 mg/kg** (then 2 mg/kg if no response; WHO), **spironolactone 1–3 mg/kg/day**, **enalapril** start 0.05–0.1 mg/kg/day titrating to ~0.5 mg/kg/day (max 20 mg), **digoxin** load 15 µg/kg then 5 µg/kg 12-hourly. Prednisolone 1–2 mg/kg/day for severe carditis (weak evidence).
23	WHO 2024 guideline: first-line drug for GAS pharyngitis and for RF/RHD prophylaxis?	**Penicillin** (IM or oral) for GAS pharyngitis; **IM BPG** for secondary prophylaxis, with **oral penicillin** as an acceptable alternative.
24	WHO 2024 on echo screening?	In moderate/high RHD prevalence settings, **echo screening (standard or handheld) of children and adolescents 5–19 years** may be implemented (strong rec); **handheld echo** can diagnose RF carditis/RHD where standard echo is unavailable.
25	Why give BPG lying down to a child with severe valve disease?	Painful injection can trigger a **vasovagal** reaction; in severe MS/AR or pulmonary hypertension this can cause collapse, often misread as anaphylaxis. Observe for 15–30 minutes.
26	Infective endocarditis prophylaxis for dental procedures (AHA) — who and what?	Only highest-risk: **prosthetic valves/material, previous endocarditis**, certain CHD. **Amoxicillin 50 mg/kg (max 2 g)** 30–60 min before; clindamycin/azithromycin if on long-term penicillin or allergic. Dental hygiene for all.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 10-year-old boy (26 kg) from Gulu has a hot, swollen, very painful right knee for 5 days, fever 38.2 °C and CRP 64 mg/L. ASO is 600 IU/mL. Echo is normal. The knee aspirate is sterile with no organisms. Which statement BEST applies the 2015 Jones criteria?

A ARF cannot be diagnosed because polyarthritis is required

B ARF cannot be diagnosed without carditis

C He has post-streptococcal reactive arthritis by definition

D He meets criteria for initial ARF: monoarthritis is a major criterion in high-risk populations, plus 2 minor (fever ≥38.0 °C and raised CRP), with GAS evidence

E He needs 3 minor criteria because this is a first episode

2. A 12-year-old girl (34 kg) is diagnosed with initial ARF with polyarthritis and moderate mitral regurgitation on echo. What is the MOST important prescription to prevent future valve damage?

A Aspirin 100 mg/kg/day for 6 months

B Benzathine penicillin G 1.2 MU IM now and every 4 weeks, for at least 10 years or until age 25 (UCG), lifelong if residual valve disease

C Penicillin V 250 mg twice daily for 10 days only

D Prednisolone 2 mg/kg/day for 3 months

E Amoxicillin 50 mg/kg before dental procedures only

3. A 9-year-old girl is brought because her handwriting has deteriorated, she drops cups, and she cries easily. On exam she has jerky movements of the face and hands, milkmaid's grip and a positive pronator sign. ASO is normal; ESR is 18 mm/h. Echo shows mild pathological MR. What is the diagnosis?

A Normal behaviour for age

B Tic disorder

C Wilson disease

D Not ARF, because ASO is normal

E Acute rheumatic fever (Sydenham chorea, with subclinical carditis)

4. A 13-year-old boy (30 kg) with known RHD (severe MR and AR) presents with orthopnoea, RR 44, HR 136, gallop, liver 5 cm and bilateral basal crackles. SpO_2 88%. BP 104/40. What is the MOST appropriate immediate management?

A Sit him up, give oxygen, IV furosemide 1 mg/kg (30 mg), restrict fluids and plan an ACE inhibitor once stable

B Normal saline 20 mL/kg bolus for the low diastolic BP

C IV propranolol for the tachycardia

D Aspirin 100 mg/kg/day

E Benzathine penicillin only and discharge

5. A 7-year-old (21 kg) with sore throat, fever 38.6 °C, tender anterior cervical nodes, tonsillar exudate and no cough presents to a HC III in Lira. No rapid strep test is available. According to WHO 2024, what is the BEST management?

A No antibiotics, because the cause cannot be confirmed
B Ciprofloxacin for 5 days
C Treat with penicillin: BPG 0.6 MU IM once (or amoxicillin 50 mg/kg once daily for 10 days)
D Wait 2 weeks and treat only if joint pains develop
E Azithromycin as first line for everyone

6. A 14-year-old girl on BPG for RHD has a documented history of an itchy rash (no swelling or breathing difficulty) after amoxicillin at age 5. She has no other allergy history. What does WHO 2024 suggest? Advanced

A Stop all prophylaxis
B Consider an oral penicillin test dose before IM BPG, as there is no history of anaphylaxis, angioedema or SJS/TEN
C Routine penicillin skin testing for every child on BPG
D Switch to ciprofloxacin prophylaxis
E Give BPG with prophylactic adrenaline every month

7. A 15-year-old with severe rheumatic mitral stenosis and pulmonary hypertension becomes pale, sweaty and hypotensive with HR 48/min two minutes after a BPG injection. There is no rash, wheeze or stridor. What is the MOST likely explanation and action? Advanced

A Anaphylaxis; give adrenaline and never give penicillin again
B Acute haemolytic reaction; send blood to the lab
C Infective endocarditis; start antibiotics
D Vasovagal reaction; lie flat with legs raised, monitor, and give subsequent injections lying down with a slower injection and pain relief
E Pulmonary embolism; start heparin

8. An 11-year-old with ARF and polyarthritis is started on aspirin 25 mg/kg 6-hourly. On day 3 she has nausea, ringing in the ears and is breathing fast with a clear chest. What is the MOST likely cause?

A Salicylate toxicity
B Worsening carditis
C Reye syndrome
D Aspiration pneumonia
E Sydenham chorea

9. School echo screening in Lira finds an asymptomatic 9-year-old (24 kg) with mild pathological mitral regurgitation meeting minimum criteria for RHD (latent RHD). No history of ARF. What is the MOST appropriate management?

A Reassure and discharge; no treatment for asymptomatic children
B Start enalapril and furosemide
C Refer for mitral valve replacement
D Repeat ASO every month and treat only if it rises
E Start BPG 0.6 MU IM every 4 weeks, enrol in the RHD clinic and repeat echo (e.g. at 2 years)

10. A 12-year-old (32 kg) had ARF with carditis 2 years ago; echo now shows no residual valve disease. According to UCG 2023, how long should BPG prophylaxis continue?

A Until 5 years after the attack only
B Until age 21 only
C For 10 years or until age 25
D For life regardless of findings
E It can stop now since the echo is normal

11. A 14-year-old with known RHD on monthly BPG has had fever for 3 weeks, weight loss, a new diastolic murmur, splinter haemorrhages, splenomegaly and microscopic haematuria. What is the MOST appropriate next step?

A Take three sets of blood cultures before starting antibiotics, and arrange echocardiography
B Start aspirin for recurrent ARF
C Give a single dose of BPG and send home
D Start antimalarials and review in a week
E Start prednisolone for carditis

12. A 10-year-old with ARF (polyarthritis, mild carditis) is on aspirin 80 mg/kg/day. After 72 hours there is no improvement in her joint pain or swelling. What is the BEST next step? Advanced

A Double the aspirin dose
B Add prednisolone immediately
C Continue the same dose for 2 more weeks before reassessing
D Reconsider the diagnosis (e.g. septic arthritis, JIA, sickle cell disease, leukaemia, post-streptococcal reactive arthritis)
E Stop BPG

13. Which auscultatory finding is MOST typical of acute rheumatic carditis in a child?

A A continuous machinery murmur below the left clavicle
B A fixed widely split second heart sound
C An apical pansystolic murmur radiating to the axilla
D An ejection systolic murmur at the upper left sternal edge with an ejection click
E A venous hum that disappears lying flat

14. A 16-year-old girl with RHD and a mechanical mitral valve on warfarin asks about contraception and future pregnancy. Which advice is MOST appropriate? Advanced

A She can stop warfarin when she becomes pregnant
B Pregnancy is low risk with a mechanical valve
C Oestrogen-containing pills are the best choice
D Warfarin is safe throughout pregnancy
E Use effective contraception (e.g. a copper IUD, implant or progestogen-only method), plan any pregnancy with the cardiologist and obstetrician, because warfarin is teratogenic in early pregnancy and mechanical valves carry high thrombosis risk

15. Which statement about the 2015 revised Jones criteria is CORRECT?

A Subclinical carditis on echo does not count as a criterion
B In moderate/high-risk populations, an ESR ≥30 mm/h counts as a minor criterion
C A prolonged PR interval counts as a minor criterion even when carditis is a major criterion
D Evidence of GAS infection is required for chorea
E Fever must be ≥39 °C to count as a minor criterion

Answers & explanations

1. D · He meets criteria for initial ARF: monoarthritis is a major criterion in high-risk populations, plus 2 minor (fever ≥38.0 °C and raised CRP), with GAS evidence

D is correct: Uganda is a moderate/high-risk setting, so **monoarthritis** counts as major; fever ≥38.0 °C and CRP ≥30 mg/L are minor; ASO provides GAS evidence: 1 major + 2 minor. A is wrong: polyarthritis is required only in low-risk populations. B is wrong: carditis is not mandatory. C is wrong: reactive arthritis is a diagnosis of exclusion and Jones criteria are met here. E is wrong: 3 minor criteria apply to recurrences, not initial episodes.

2. B · Benzathine penicillin G 1.2 MU IM now and every 4 weeks, for at least 10 years or until age 25 (UCG), lifelong if residual valve disease

B is correct: **secondary prophylaxis with BPG** every 4 weeks prevents recurrences that worsen RHD; at ≥30 kg the dose is 1.2 MU, and with carditis the duration is long (residual disease: until 40–45 or life). A is wrong: aspirin treats arthritis but does not prevent RHD, and 6 months is excessive. C is wrong: 10 days of penicillin only eradicates current GAS. D is wrong: steroids have weak evidence and no effect on recurrence. E is wrong: endocarditis prophylaxis does not prevent ARF recurrence.

3. E · Acute rheumatic fever (Sydenham chorea, with subclinical carditis)

E is correct: **Sydenham chorea is diagnostic of ARF on its own**, and appears 1–6 months after GAS when ASO may have normalised; the echo also shows subclinical carditis, which lengthens prophylaxis. A is wrong: these are neurological signs. B is wrong: tics are stereotyped and suppressible, without milkmaid's grip or hypotonia. C is wrong: Wilson disease is rare at this age and causes liver disease and Kayser–Fleischer rings. D is wrong: GAS evidence is not required for chorea.

4. A · Sit him up, give oxygen, IV furosemide 1 mg/kg (30 mg), restrict fluids and plan an ACE inhibitor once stable

A is correct: he has acute **heart failure** from regurgitant lesions; sit up, oxygen to SpO_2 >90%, IV furosemide (WHO 1 mg/kg, then 2 mg/kg if no response), restrict fluids, then afterload reduction with an ACE inhibitor for MR/AR. B is wrong: the low diastolic BP is from aortic regurgitation; fluid would worsen pulmonary oedema. C is wrong: the tachycardia is compensatory; IV beta-blockade can cause collapse. D is wrong: there is no evidence of active ARF arthritis here. E is wrong: BPG is essential but does not treat acute failure, and he needs admission.

5. C · Treat with penicillin: BPG 0.6 MU IM once (or amoxicillin 50 mg/kg once daily for 10 days)

C is correct: WHO 2024 recommends that in **moderate/high-risk settings without testing**, children with clinically suspected GAS pharyngitis receive antibiotics, with **penicillin (IM or oral) first line**; under 30 kg the BPG dose is 0.6 MU. A is wrong: that misses primary prevention. B is wrong: ciprofloxacin has poor GAS activity. D is wrong: treating after ARF starts does not prevent it. E is wrong: macrolides are for penicillin allergy; resistance is a concern.

6. B · Consider an oral penicillin test dose before IM BPG, as there is no history of anaphylaxis, angioedema or SJS/TEN

B is correct: WHO 2024 conditionally recommends an **oral penicillin test dose** before BPG in patients with a history of **mild** penicillin allergy (no anaphylaxis, angioedema or SJS/TEN). A is wrong: stopping prophylaxis exposes her to recurrence. C is wrong: WHO advises against allergy testing in patients **without** an allergy history, and testing is not routine for all. D is wrong: ciprofloxacin is not an anti-streptococcal prophylactic. E is wrong: prophylactic adrenaline is not indicated.

7. D · Vasovagal reaction; lie flat with legs raised, monitor, and give subsequent injections lying down with a slower injection and pain relief

D is correct: **bradycardia with hypotension and pallor without rash, wheeze or stridor** after a painful injection is a **vasovagal** reaction, which can be dangerous in severe valve disease; manage supine with legs raised, and prevent next time (lying down, lidocaine diluent, slow injection). A is wrong: anaphylaxis typically causes tachycardia with skin or airway signs (though if in doubt, adrenaline is not withheld). B is wrong: no blood was transfused. C is wrong: endocarditis is subacute, not a 2-minute event. E is wrong: nothing suggests PE and heparin is not indicated.

8. A · Salicylate toxicity

A is correct: **tinnitus, nausea and hyperventilation** (respiratory stimulation) are classic **salicylate toxicity**; reduce or stop the dose, check a salicylate level if possible, or switch to naproxen. B is wrong: carditis would cause tachycardia, a gallop and crackles, not tinnitus. C is wrong: Reye syndrome causes encephalopathy and liver failure, usually after a viral illness. D is wrong: the chest is clear. E is wrong: chorea causes movements, not tinnitus or tachypnoea.

9. E · Start BPG 0.6 MU IM every 4 weeks, enrol in the RHD clinic and repeat echo (e.g. at 2 years)

E is correct: WHO 2024 (strong recommendation, based on **GOAL** in Ugandan children) advises **secondary prophylaxis for children with screen-detected RHD**, which reduced progression from 8.2% to 0.8%. A is wrong: untreated latent RHD can progress. B is wrong: she has no heart failure. C is wrong: mild MR does not need surgery. D is wrong: ASO monitoring does not prevent recurrence.

10. C · For 10 years or until age 25

C is correct: UCG 2023 recommends **10 years or until age 25** for carditis without residual disease (AHA: 10 years or age 21, whichever is longer). A is wrong: 5 years is for ARF without carditis. B is wrong: age 21 alone applies to the no-carditis category (UCG 18–21). D is wrong: lifelong is for residual valve disease. E is wrong: stopping early risks recurrence.

11. A · Take three sets of blood cultures before starting antibiotics, and arrange echocardiography

A is correct: fever, a new murmur, splinters, splenomegaly and haematuria in RHD suggest **infective endocarditis**; take **three blood cultures before antibiotics** and do echo, then give 4–6 weeks of IV antibiotics. B is wrong: ARF recurrence does not explain embolic phenomena. C is wrong: BPG prophylaxis does not treat endocarditis. D is wrong: a 3-week fever with these signs needs cultures; malaria testing can be added but is not the answer. E is wrong: steroids would worsen infection.

12. D · Reconsider the diagnosis (e.g. septic arthritis, JIA, sickle cell disease, leukaemia, post-streptococcal reactive arthritis)

D is correct: ARF arthritis typically responds **dramatically within 24–72 hours** to salicylates or NSAIDs; failure to respond should prompt reconsidering the diagnosis. A is wrong: doubling risks salicylate toxicity. B is wrong: steroids do not fix a wrong diagnosis and may hide leukaemia. C is wrong: waiting delays an alternative diagnosis such as septic arthritis. E is wrong: BPG prophylaxis is unrelated to the joint response.

13. C · An apical pansystolic murmur radiating to the axilla

C is correct: **mitral regurgitation** (apical pansystolic murmur radiating to the axilla) is the commonest finding in acute rheumatic carditis. A is wrong: that is a patent ductus arteriosus. B is wrong: a fixed split S2 suggests an atrial septal defect. D is wrong: that suggests pulmonary valve stenosis. E is wrong: a venous hum is an innocent murmur.

14. E · Use effective contraception (e.g. a copper IUD, implant or progestogen-only method), plan any pregnancy with the cardiologist and obstetrician, because warfarin is teratogenic in early pregnancy and mechanical valves carry high thrombosis risk

E is correct: mechanical valves need anticoagulation throughout pregnancy and pregnancy is high risk; **warfarin is teratogenic** (especially weeks 6–12), so pregnancy must be planned with specialists, and reliable non-oestrogen contraception is preferred. A is wrong: stopping anticoagulation risks valve thrombosis. B is wrong: risk is high. C is wrong: oestrogen increases thrombosis risk. D is wrong: warfarin embryopathy occurs in the first trimester.

15. B · In moderate/high-risk populations, an ESR ≥30 mm/h counts as a minor criterion

B is correct: high-risk thresholds are **ESR ≥30 mm/h and/or CRP ≥3.0 mg/dL**, with fever ≥38.0 °C. A is wrong: subclinical (echo-detected) carditis counts as major since 2015. C is wrong: PR prolongation cannot be counted when carditis is already a major criterion. D is wrong: chorea alone is enough. E is wrong: the fever threshold is ≥38.0 °C (high risk) or ≥38.5 °C (low risk).

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The boarding-school girl with jumping joints

Atim, 12 years, a boarder at a primary school in Lira, is brought to the regional referral hospital with 10 days of painful joints and fever. She was treated at the school sick bay with paracetamol.

1. What history will help you most?

> **Pattern of joint involvement** (migratory, large joints, very painful), **sore throat 2–4 weeks ago**, skin sores, fever, chest pain/breathlessness/orthopnoea, abnormal movements or emotional change, rash/lumps, previous ARF or injections, **penicillin allergy**, family history, crowding, nutrition, immunisation, HIV status, menarche.

2. History: right knee then left ankle then right wrist, sore throat 3 weeks ago untreated; mild exertional breathlessness. Exam: 31 kg, T 38.4 °C, HR 120 (and 112 asleep), swollen tender left ankle, apical 3/6 pansystolic murmur to the axilla, no gallop, liver not enlarged. What criteria does she meet?

> **Major**: polyarthritis (migratory) and **carditis** (MR murmur, tachycardia out of proportion). **Minor**: fever ≥38.0 °C, plus ESR/CRP once available. With GAS evidence (ASO), this is **initial ARF with carditis**.

3. What investigations do you request and what do you expect?

> **ESR/CRP** (raised), **ASO** ± anti-DNase B (raised), **throat swab** (may be negative), **FBC**, **ECG** (PR may be long, but it cannot count as minor because carditis is major), **CXR** (heart size, congestion), **echo** (pathological MR per WHF criteria; check for MS, AR, effusion, LV function).

4. ESR 92, ASO 720, echo moderate MR with thickened leaflet tips, LV mildly dilated, good function. Write her management.

> Admit; rest while active then mobilise. **BPG 1.2 MU deep IM today** (≥30 kg) to eradicate GAS and start prophylaxis. **Aspirin 80–100 mg/kg/day in 3–4 doses** (≈2.5–3 g/day; max ~4 g) or **naproxen 10–20 mg/kg/day in 2 doses (max 1 g)** until joints settle and CRP falls, then taper. Monitor for heart failure (weight, RR, HR, liver). Consider enalapril if LV dilatation progresses. Register in the RHD clinic.

5. How long will she need BPG and what will you tell her family?

> **Every 4 weeks** for at least **10 years or until age 25** (UCG), and **until 40–45 or for life** if valve disease persists on follow-up echo. Explain that injections stop further attacks that damage the heart; missing them is dangerous; pain can be reduced; she must come early for breathlessness or swelling; good dental care; future pregnancies must be planned with a doctor.

6. What would make you suspect an alternative diagnosis during follow-up?

> **No response of arthritis within 72 hours** of salicylates/NSAIDs (septic arthritis, JIA, SCD, leukaemia), persistent fever with new murmur and embolic signs (**endocarditis**), cytopenias or organomegaly (leukaemia).

Take-away: Migratory polyarthritis with a new MR murmur after a sore throat is ARF with carditis. Apply high-risk Jones criteria, give BPG on day one, treat inflammation, echo everyone, and commit the family to years of 4-weekly injections.

Case 2: Breathless at 13: established RHD in heart failure

Okot, a 13-year-old boy (29 kg), is carried into the paediatric ward at night unable to lie flat. He was diagnosed with 'heart disease' 2 years ago and was supposed to get monthly injections but stopped after 4 because the health centre ran out.

1. What are your first actions?

> **Sit him up**, **oxygen** to SpO_2 >90%, IV access, vitals (HR, rhythm, BP, RR, SpO_2, temperature), **HemoCue Hb**, glucose. Look for **gallop, crackles, liver size, JVP, oedema**. Avoid IV fluid boluses.

2. Findings: RR 48, SpO_2 87%, HR 140 irregularly irregular, BP 96/58, gallop, liver 6 cm, bilateral crackles, apical PSM + mid-diastolic rumble with loud S1. Hb 10.5 g/dL, T 37.4 °C. What is going on?

> **Acute decompensated heart failure** from **chronic RHD with mixed mitral valve disease (MR + MS)**, precipitated by **new atrial fibrillation** and possibly **ARF recurrence** (prophylaxis stopped). Also consider endocarditis and infection.

3. Write the immediate treatment.

> **Oxygen**, sit up. **Furosemide 1 mg/kg IV (≈30 mg)**; if no diuresis in 2 h, **2 mg/kg**. **Rate control for AF**: **digoxin** oral load 15 µg/kg (WHO; ≈435 µg, often split into two doses 6 h apart) then 5 µg/kg 12-hourly (max 250 µg/dose), or a beta-blocker when not in acute failure. Check K^+ (avoid hypokalaemia with digoxin). Fluid and salt restriction. ECG, CXR, ESR/CRP, ASO, blood cultures if febrile. **Anticoagulation** for AF with MS (heparin then warfarin) after specialist discussion.

4. Investigations show ESR 70, ASO 560, CRP 48. Echo: moderate MS (mitral valve area reduced), moderate MR, left atrium dilated, pulmonary artery pressure raised. How does this change your plan?

> Likely **recurrent ARF** on chronic RHD. Restart **BPG every 4 weeks** (at 29 kg: **0.6 MU** by the UCG 30 kg cut-off; AHA would give 1.2 MU above 27 kg; follow your clinic protocol and re-weigh when euvolaemic), treat inflammation (aspirin or naproxen) if arthritis/active carditis, continue diuretics and rate control. Refer to the **Uganda Heart Institute** for assessment for **balloon mitral valvotomy or surgery**.

5. Why should you be cautious with ACE inhibitors here?

> In **stenosis-dominant** disease, cardiac output depends on filling across a narrowed valve; vasodilators can drop BP without relieving the obstruction. ACE inhibitors are most useful for **regurgitant** lesions. Rate control (longer diastole) and diuretics help MS; definitive treatment is mechanical.

6. How will you prevent this happening again?

> Register in an **RHD clinic**, provide a **BPG card** and schedule close to home, use pain-reduction methods, contact the community health worker for **tracing defaulters**, educate the family and school, report **BPG stock-outs** and arrange alternative supply (oral penicillin V 250 mg twice daily as a fallback), and plan follow-up echo and anticoagulation monitoring.

Take-away: Stopping prophylaxis leads to recurrence and decompensation. Treat failure and AF, restart BPG, refer stenosis for mechanical relief, and build adherence support around the family.

Case 3: The clumsy girl

Nansubuga, a 9-year-old girl (23 kg) from Mukono, is brought because her teacher reports she has become 'restless and careless', cannot write neatly, and laughs and cries easily. Her mother noticed she drops cups.

1. What examination findings will you look for?

> **Chorea**: involuntary purposeless movements of face and limbs, **milkmaid's grip**, **darting tongue**, **pronator sign**, **spooning**, hypotonia, difficulty with a spiral drawing; check the movements disappear in sleep. Full **cardiovascular exam** (murmurs, tachycardia), joints, skin (nodules, erythema marginatum), growth, BP.

2. She has milkmaid's grip and a positive pronator sign. Heart sounds normal, no murmur. ESR 20, ASO 150 IU/mL (normal). Can you diagnose ARF?

> **Yes**: **Sydenham chorea alone** diagnoses ARF, without other criteria or GAS evidence, because chorea appears 1–6 months after infection when ASO and inflammatory markers may have normalised. Exclude other causes (drugs, SLE, Wilson, hyperthyroidism) clinically.

3. What investigation is still essential?

> **Echocardiography**: subclinical carditis is common with chorea and determines the **duration of prophylaxis**. (Hers shows mild pathological MR: subclinical carditis.)

4. Write her treatment.

> **BPG 0.6 MU IM** (<30 kg) now and **every 4 weeks**. **Chorea**: calm environment, explain to school; if disabling, **sodium valproate 10–20 mg/kg/day** in 2–3 doses (≈115–230 mg 12-hourly) or carbamazepine 7–20 mg/kg/day. No aspirin needed (no arthritis). Prophylaxis duration: carditis present → at least **10 years or until 25** (UCG), longer if residual disease on follow-up echo.

5. How long will the chorea last and what should you tell the teacher?

> Usually **weeks to months** (typically resolves within 6 months, sometimes recurs). Explain it is a **medical illness, not misbehaviour**; allow extra time for written work, avoid punishment, and reduce stress.

6. Present her on the ward round in one paragraph.

> "Nansubuga, 9 years, 23 kg (WAZ −0.8), immunised per UNEPI, HIV-negative, presents with 6 weeks of involuntary movements, clumsiness and emotional lability; exam shows Sydenham chorea with no murmur; ASO and ESR normal; echo shows mild pathological MR (subclinical carditis). Diagnosis: ARF (Sydenham chorea) with subclinical carditis. Plan: BPG 0.6 MU 4-weekly for at least 10 years or until 25, valproate for disabling chorea, school liaison, repeat echo in 1 year."

Take-away: Chorea is enough to diagnose ARF even with normal ASO. Always echo, because subclinical carditis decides how long BPG must continue.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.